Intoxicated Speech Detection

CL Team Lab Project Presentation

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Summer Semester 2023

April 22, 2026

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Outline

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Introduction

Motivation

The **task** of intoxicated speech detection is in the area of paralinguistic speech research and allows to predict the level of alcohol in blood/breath based on suprasegmental information in recorded speech utterances.

- *Possible applications*: Preventing alcohol-related incidents, safety at workplace (automatic tests where alertness and coordination are required), alcohol abuse monitoring and medical treatment.
- *Challenges*: speaker and context variability $\longrightarrow$ hard to build a robust model
- *Human results* from a perceptual listening experiment: **accuracy =** 73.8% (Mullennix et al., 1989)

Empirical Observations

Tendencies observed despite individual speaker variation:

- Intoxicated speech is subject to these alternations: [s] $\longrightarrow$ [ʃ] , [t] $\longrightarrow$ [k], plosives $\longrightarrow$ their vocal counterparts (Schiel et al., 2008).
- Disfluency (break/irregularity) rates, as well as durations, increase with intoxication in spontaneous speech and most reading, except simple commands (Schiel and Heinrich, 2015).
- Intoxicated speech contains less repetitions (Schiel and Heinrich, 2015)
- Gender does not play a role (Schiel and Heinrich, 2015)

Data

ALC Corpus

Short for **Alcohol Language Corpus** (Schiel et al., 2008)
$\Longrightarrow$ a corpus including intoxicated (as well as sober) speech

- 162 speakers
- 14,580 recordings
- different type of prompts e.g. tongue-twisters, numbers, monologue etc.
- participants were recorded in both a sober and an intoxicated state
- includes recordings, useful metadata, as well as transcriptions

ALC Corpus: Details

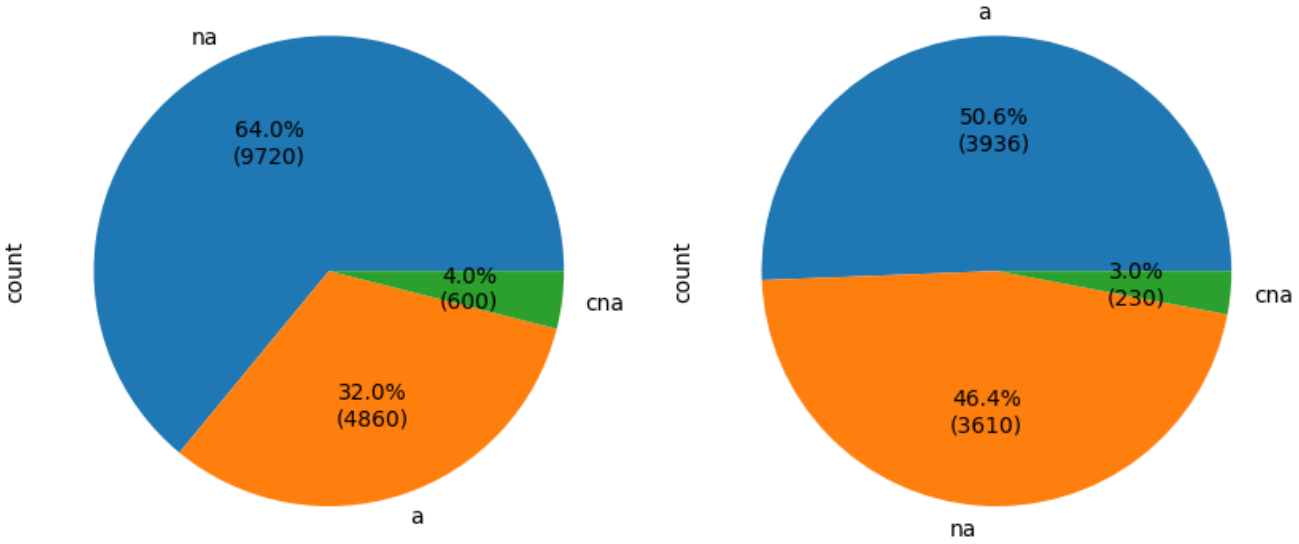
The content and task type do not constitute a hierarchy: one content may belong to different types and vice versa.

- **Type**: R/E/M/D/L = Read / Elicited / Monologue / Dialogue / List
- **Content**: A/P/Q/N/R/C/S/T = Address / Picture / Question / Number / Read Command / Spont. Command / Spelling / Tongue twister

type	D	D	E	L	L	L	M	M	R	R	R
cont.	P	Q	C	N	S	T	P	Q	A	R	T
#	506	668	2530	2530	344	344	1012	344	2530	2186	2186

Methods

Balancing Data



Original dataset (on the left) and one of the smaller balanced subsets (on the right).

- Original (**full**) dataset was split into train/test/validation partitions randomly.
- At the same time, a smaller subset balancing intoxicated and non-intoxicated classes randomly was created, further referred to **'50-50 split'**.
- Next, smaller datasets were created to balance the number of datapoints per speaker. Document ids already in *full* test/validation were excluded and new small subset rebalanced. Two separate subsets were created:
  - aiming to balance for **speaker + task content**
  - aiming to balance for **speaker + task type**

Global Speaker Normalisation

Global Speaker Normalisation (Bone et al., 2014):

- normalise the feature values for each speaker across all data points from that speaker
- use Z-normalisation

$$New_value = (x - \mu)/\sigma$$

x: Original value
μ: Mean of data
σ: Standard deviation of data

Models

- Functional Features
  - ML Baselines (sklearn)
  - Simple Neural Network (PyTorch)
- LLD Features
  - LSTM (PyTorch)
  - Transformer (PyTorch)

Experiments & Results

Baseline (Functional)

Dataset Evaluation	full 0	50/50 1	50/50n 1	s-type 0	s-type(n) 1	s-cont 0	s-cont(n) 1
Majority	68.906	45.165	45.165	68.906	54.835	68.906	54.835
Naive Bayes	65.876	58.539	63.374	60.738	62.963	58.564	62.963
kNN	63.834	55.761	63.272	54.348	61.420	55.731	61.420
SVM (linear)	69.236	55.350	66.152	32.082	63.786	31.225	63.889
SVM (poly)	68.906	61.626	66.152	55.731	**65.741**	68.314	65.741
LogReg	70.026	61.111	66.255	62.516	63.786	61.989	63.786
RandForest	**74.967**	70.782	68.519	69.565	65.329	72.925	**66.358**

Evaluation strategies: (0) original (full) validate and test (1) internal validate and test (part of subset). Datasets: (s-type): balanced for speaker and task types (but not content), (s-cont): balanced for speaker and task content (but not type).

Hyperparameter Search (Functional)

Combo	Epochs	B. size	L.rate	Layers	Optim.	Droupout
combo 0	10	5	0.0001	[64, 32, 16, 8, 4]	Adam	False
combo 1	10	5	0.0001	[32]	Adam	False
combo 2	10	5	0.0001	[64, 16, 4]	Adam	False
combo 3	10	10	0.0001	[64, 32, 16, 8, 4]	Adam	False
combo 4	10	5	0.0001	[64, 32, 16, 8, 4]	Adam	0.5
combo 5	10	10	0.0001	[32]	Adam	False
combo 6	10	5	0.0001	[64, 32, 16, 8, 4]	SGD	False
combo 7	20	10	0.0001	[32]	Adam	False
combo 8	10	5	0.00001	[64, 32, 16, 8, 4]	Adam	False
combo 9	10	5	0.0005	[64, 32, 16, 8, 4]	Adam	False
combo 10	10	15	0.0001	[32]	Adam	False
combo 11	10	15	0.0001	[32]	Adam	0.1

Functional Feature Results

Param. combo	50/50 split
combo 0	60.185
combo 1	60.905
combo 2	61.008
combo 3	**62.654**
combo 4	45.165
combo 5	59.671
combo 6	54.012
combo 7	61.934
combo 8	58.539
combo 9	59.362
combo 10	60.802
combo 11	59.774

Accuracy of the simple neural network variantions on the 50/50 data split.

Functional Feature Results

Param. combo	50/50 split	50/50 split (norm.)
combo 0	60.185	66.152
combo 1	60.905	66.77
combo 2	61.008	64.609
combo 3	**62.654**	66.255
combo 4	45.165	45.165
combo 5	59.671	67.078
combo 6	54.012	65.844
combo 7	61.934	66.461
combo 8	58.539	66.358
combo 9	59.362	63.169
combo 10	60.802	66.872
combo 11	59.774	**68.107**

Accuracy of the simple neural network variantions on the 50/50 data split.

Functional Feature Results

Param. combo	50/50 split	50/50 split (norm.)
combo 0	60.185	66.152 (+5.97)
combo 1	60.905	66.77 (+5.86)
combo 2	61.008	64.609 (+3.6)
combo 3	**62.654**	66.255 (+3.6)
combo 4	45.165	45.165 (+0.0)
combo 5	59.671	67.078 (+7.41)
combo 6	54.012	65.844 (**+11.83**)
combo 7	61.934	66.461 (+4.53)
combo 8	58.539	66.358 (+7.82)
combo 9	59.362	63.169 (+3.81)
combo 10	60.802	66.872 (+6.07)
combo 11	59.774	**68.107** (+8.33)

Accuracy of the simple neural network variantions on the 50/50 data split.

Functional Feature Results

Param. combo	50/50 split	50/50 split (norm.)
combo 0	60.185	66.152 (+5.97)
combo 1	60.905	66.77 (+5.86)
combo 2	61.008	64.609 (+3.6)
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combo 8	58.539	66.358 (+7.82)
combo 9	59.362	63.169 (+3.81)
combo 10	60.802	66.872 (+6.07)
combo 11	59.774	**68.107** (+8.33)

Accuracy of the simple neural network variantions on the 50/50 data split.

$\Longrightarrow$ average improvement of **5.74 points**

Functional Feature Results

Combo	bal. for C	bal. for C (norm.)	bal. for T	bal. for T (norm.)
combo 0	57.407	63.477	54.835	60.391
combo 1	44.959	64.918	44.753	63.066
combo 2	45.165	62.346	49.486	61.008
combo 3	46.605	63.169	50.206	60.185
combo 4	45.165	54.835	54.835	53.909
combo 5	58.745	63.889	58.23	62.551
combo 6	54.733	62.551	51.029	61.42
combo 7	45.576	62.963	45.062	62.243
combo 8	57.51	62.243	53.086	62.346
combo 9	47.119	**65.021**	55.247	64.403
combo 10	**59.156**	64.095	**59.053**	**64.918**
combo 11	54.835	63.477	57.716	63.066

Accuracy of the SNN variantions on the balanced 50/50 data splits.

$\Longrightarrow$ average improvement of **11.33** / **8.83 points**

Hyperparameter Search (LLD)

Combo	B. size	Layers	Droupout	Bias	Bidir.
combo 0	5	[16, 8, 4]	False	True	False
combo 1	5	[8]	False	True	False
combo 2	5	[16, 8, 4]	False	True	True
combo 3	5	[16, 8, 4]	False	False	False
combo 4	10	[16, 8, 4]	False	True	False
combo 5	5	[16, 8, 4]	0.5	True	False

LLD Features

A simple transformer model from scratch (dropout=0.2 at encoder, 10 epochs).

- Learned on balanced data, applied on original full validation subset.
- Accuracy around 48.87%.
- Analysis: transformer architecture benefits from quick processing of large amounts of data. Training on a balanced but small dataset produces worse results.
- Future work: adapt a pretrained transformer from HF based on different (not OpenSmile LLD) features.

LLD Feature Results

Combo	bal.	bal. (norm.)	bal. for C	bal. for C (norm.)	bal. for T	bal. for T (norm.)
combo 0	46.811	54.835	51.646	**54.63**	**51.646**	**52.778**
combo 1	46.193	52.058	48.971	52.16	50.206	50.309
combo 2	**53.601**	52.778	50.926	50.926	48.868	50.206
combo 3	52.469	52.778	**52.881**	49.177	50.206	51.955
combo 4	51.337	**59.156**	52.366	48.971	50.514	50.617
combo 5	48.868	48.868	49.486	51.132	48.765	51.132

Accuracy of the LSTM variantions on the 50/50 data splits.

$\Longrightarrow$ average improvement of **3.53** / **0.12** / **1.13 points**

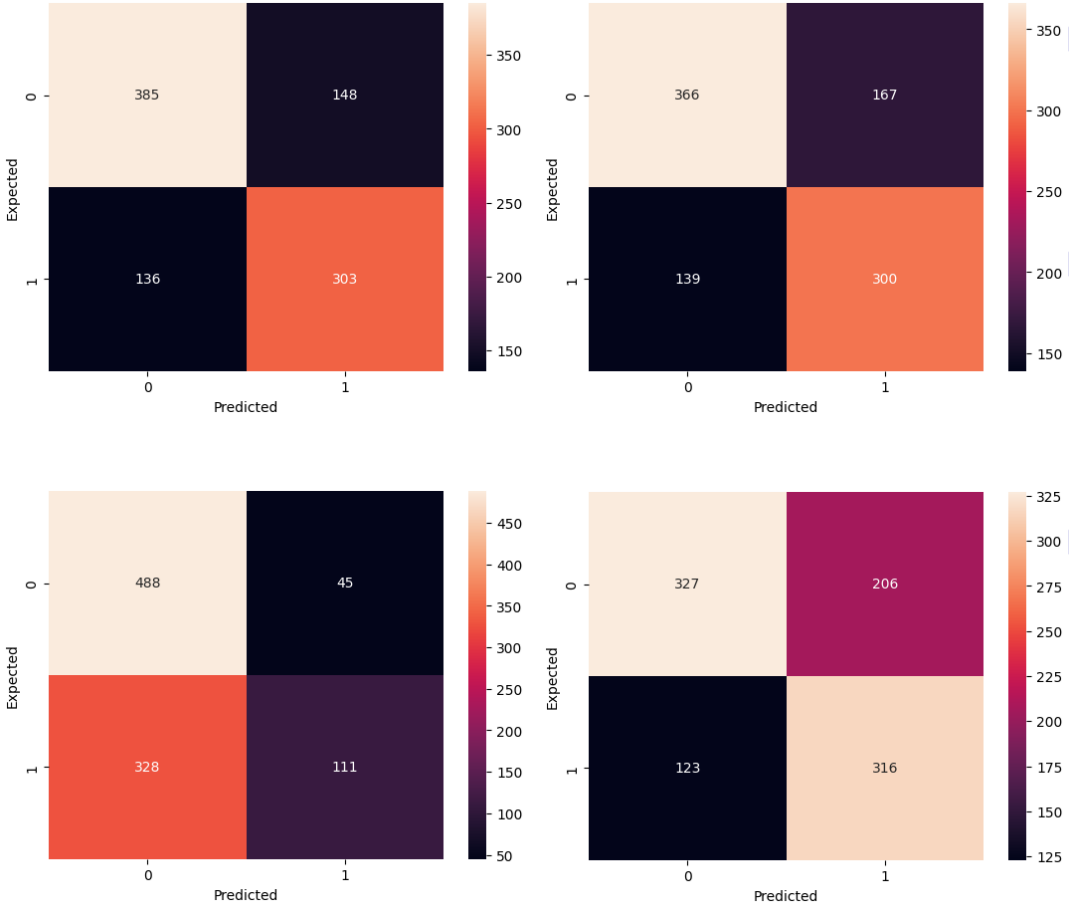
Conclusions from Hyperparameter Search

- Dropout of 0.5 decreased performance
- A small number of linear layers can already be sufficient
- Normalising the data often had an effect on the best architecture

Analysis

Analysis Functional (Baseline)

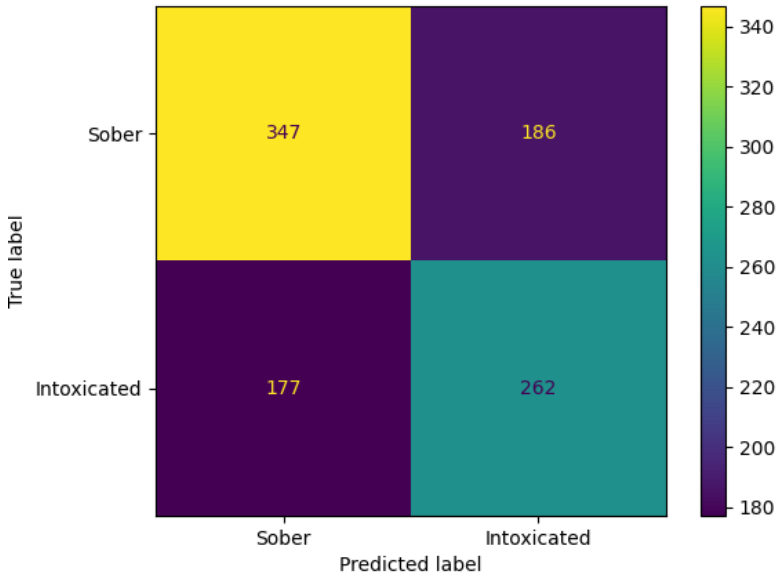
Comparison of the best (RF) model and a strong baseline SVM (Poly) model with and without normalisation on the 50-50 subset.



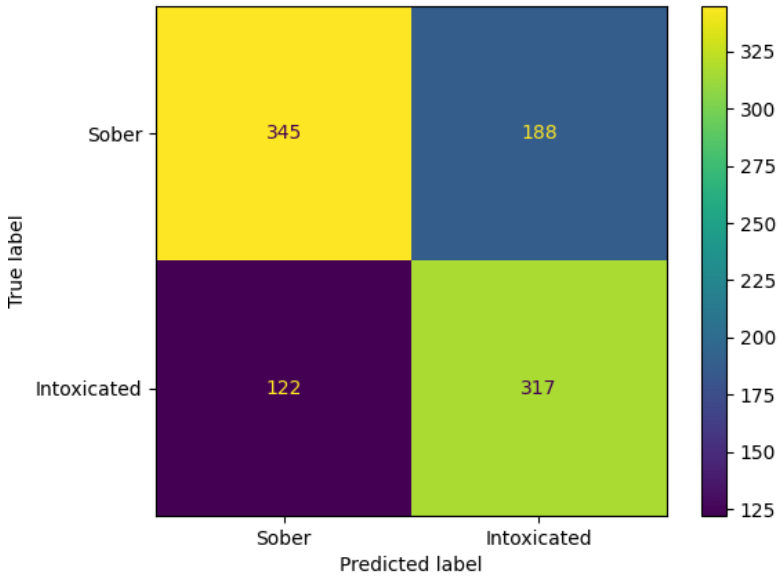
- Most models showed a **5-10%** improvement on normalized data vs non-normalized.
- Random Forest was the only model that did not significantly benefit from normalization, or even performed worse.
- This is probably due to the fact that RF finds a particular feature and sticks to it for classification, while other baseline models rely on the whole vector representation.

Analysis Functional

Best model w/o normalisation (on the 50/50 split):



Best model with normalisation (on the 50/50 split):



Analysis Functional

Prompts:

	not norm.	norm.
Read	63.19%	69.63%
Free	61.56%	65%

50/50 split

	not norm.	norm.
Read	58.9%	65.49%
Free	59.69%	64.06%

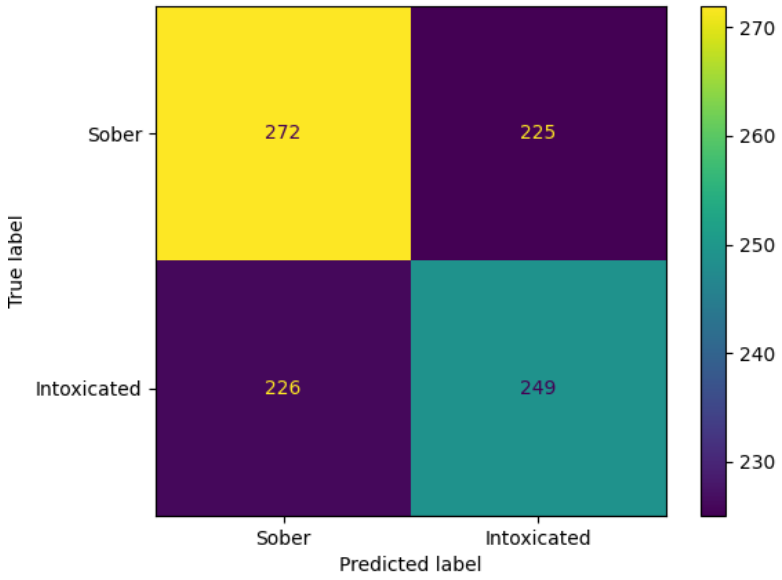
50/50 split, balanced (C)

	not norm.	norm.
Read	57.82%	65.18%
Free	61.56%	64.38%

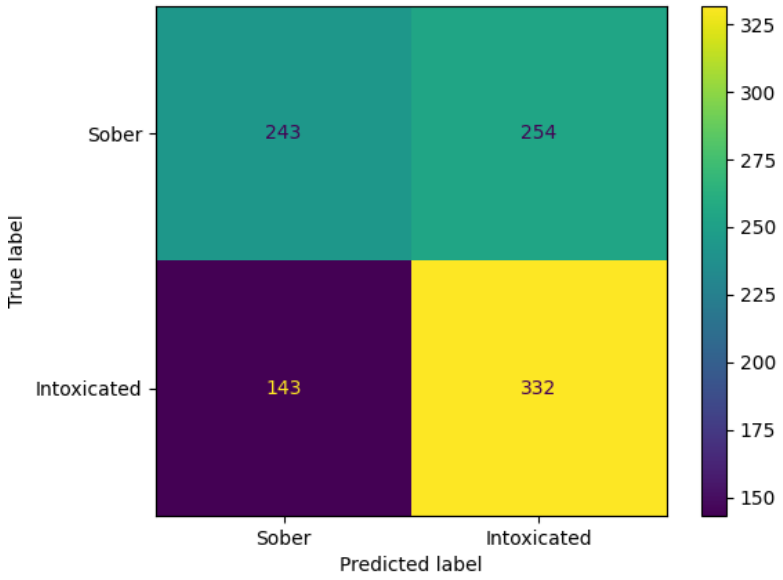
50/50 split, balanced (T)

Analysis LLD

Best model w/o normalisation (on the 50/50 split):



Best model with normalisation (on the 50/50 split):



Analysis LLD

Prompts:

	not norm.	norm.
Read	52.4%	58.58%
Free	56%	60.31%

50/50 split

	not norm.	norm.
Read	53.63%	54.71%
Free	51.38%	54.46%

50/50 split, balanced (C)

	not norm.	norm.
Read	52.09%	53.79%
Free	50.77%	50.77%

50/50 split, balanced (T)

Achievements & Project Planning

Project Planning

Week 1:

- Research (model architectures & task)
- Data analysis & preprocessing

Week 2:

- **Feature extraction**
- **Baselines**
- Evaluation

Week 3 & 4:

- Code DL models
- Analyse baselines

Project Planning

Week 5:

- **Finish coding models & hyperparameter experiments**
- Documentation & analysis of results

Week 6 & 7:

- **Find & extract interesting features**
- Conduct ablation studies
- Run experiments on more “advanced” datasets

Week 8:

- Documentation & analysis of results
- Preparation of presentation

Project Planning

Week 5:

- **Finish coding models & hyperparameter experiments**
- Documentation & analysis of results

Week 6 & 7:

- ~~**Find & extract interesting features**~~ **GSN & subset creation**
- Conduct ablation studies
- Run experiments on more “advanced” datasets

Week 8:

- Documentation & analysis of results
- Preparation of presentation

Project Planning

What went well:

- good plan with enough buffers
- weekly progress

What we should consider in the future:

- conduct literature review **before** planning the project
- maybe more regular meetings

What's still missing

- train the (best) models on the unbalanced dataset
- merge the two best performing models (Functionals + LLDs)

References

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